

Direct Action Planning: An Auditable Action-Comparison Contract for Finite-Budget Resource Scheduling

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Finite-budget resource schedulers allocate capacity under uncertain demand while reserving resources for future work. Making these decisions auditable requires a common interface between candidate evaluation, action selection, and resource accounting. We present Direct Action Planning (DAP), an action-comparison contract that carries a decision's rationale through execution. DAP constructs explicit action effects under one causal forecast, evaluates the continuation value of affordable post-cost states, and sends the feasible score maximizer directly to the actuator. Its decision record links candidate consequences, resource commitments, scores, and execution receipts, enabling independent checks across software components. We evaluate DAP on two public workload traces, an exhaustive dynamic-programming diagnostic, executable fault injections, and a Kubernetes implementation. On GenTD26, explicit action effects raise completion by 3.79 percentage points and save 19.09 resource units relative to a matched learned-transition control. The checker detects and localizes all 3,554 activated contract violations across two value representations, with zero false positives in 640 clean contexts and a median checking cost of 7.80 microseconds. In 160 matched Kubernetes runs, DAP delivers every logged feasible maximizer; on Azure, it raises completion by 6.77 points and reduces Ready-resource cost by 54.4% relative to immediate scoring. These results show how explicit action semantics can support efficient resource allocation and independently checkable execution within one scheduling interface.

CCS Concepts: • **Computing methodologies** → **Planning and scheduling**; • **Software and its engineering** → **Software performance**.

Additional Key Words and Phrases: resource scheduling, finite budget, approximate dynamic programming, action-comparable planning, Kubernetes, autoscaling

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1 Introduction

Resource scheduling connects software decisions to capacity. In autoscaling, admission control, and queue management, a controller chooses a concrete command while preserving resources for future demand. Capacity used to clear a queue now is unavailable for a later burst. Finite-budget scheduling therefore requires both an estimate of future opportunity and an explicit account of what each affordable action changes. Figure 1 illustrates two candidate consequences: serving more work now consumes future opportunity, whereas waiting preserves resource while accepting queue pressure. The scheduler compares these consequences under the same remaining budget and horizon.

Deployment adds a second requirement: the choice must remain identifiable across forecasting, value estimation, actuation, and accounting. Consider a controller choosing among absolute replica targets. Two targets may both satisfy a local API check and the budget, yet only one maximizes the planner's score. Verifying execution requires connecting those scores to the delivered command and its resource charge. An action-level record makes that relationship observable throughout the control cycle.

Existing approaches provide complementary foundations. Model-free policies learn long-run choices from interaction [37, 42, 46], and constrained reinforcement learning (RL) incorporates resource costs and policy constraints [1, 12, 40, 43, 52]. Model predictive control (MPC) exposes

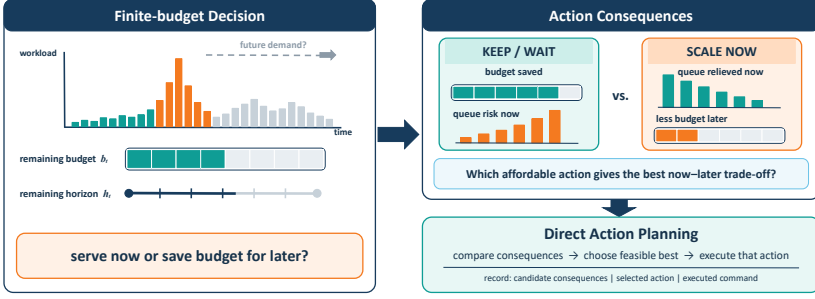


Fig. 1. Motivation for direct action planning. The schematic endpoints show how a lower-resource action preserves later opportunity, while a higher-resource action can improve current service at the cost of future budget. DAP compares the affordable consequences among four capacity levels and executes the best now-later trade-off.

model structure through online lookahead [31]; post-decision-state approximate dynamic programming (PDS/ADP) separates controlled effects from subsequent exogenous uncertainty [8, 25, 35]. Our focus is their connection to execution: candidate consequences, affordability, scores, and actuator identity must describe the same choice. Resource effects available as equations or calibrated API semantics provide a basis for making that connection explicit.

We introduce Direct Action Planning (DAP), an action-comparison contract for finite-budget scheduling. Its central idea is to make the feasible action comparison an explicit software object. Every candidate receives one common causal forecast; its known resource effect constructs a post-cost state; and a learned continuation value estimates the opportunity remaining after that action. The feasible maximizer is sent directly to the actuator. The record binds these computations to the delivered command and measured ledger update, enabling a checker to reconstruct the selection and localize a disagreement at a software boundary.

Explicit effects preserve action semantics, while continuation value amortizes longer-horizon reasoning into one-step comparisons. A PDS adapter reuses the record with a different value representation. Two public traces, exact dynamic programming (DP), fault injections, and Kubernetes runs evaluate these capabilities: matched controls test service productivity and action ordering, and independent receipts verify delivery.

The paper makes three contributions:

- (1) **An action-comparison contract for resource scheduling.** We formalize a decision record that connects causal context, explicit candidate effects, external affordability, action scores, and execution identity. The contract makes feasibility, service productivity, and planner-to-actuator correctness separately measurable (Section 2).
- (2) **Efficient planning with explicit resource effects.** We develop a structured planner that evaluates affordable post-cost states with a shared forecast and budget-conditioned continuation. Matched trace controls and exact-DP diagnostics establish its action-ordering benefits, alongside sub-millisecond mean planning in the matched evaluation (Sections 3 and 5.5).
- (3) **Reusable verification and a deployed realization.** Cross-boundary checks support both DAP and PDS. A Kubernetes controller realizes the contract through direct actuation and independent Ready-time accounting. Fault injections test detection and localization; paired live runs demonstrate service-resource gains and command fidelity (Sections 4 and 5.6).

2 Problem and Design Requirements

2.1 Finite-Budget Resource Scheduling

At time t , a controller observes causal system state s_t , remaining budget b_t , and remaining horizon h_t . Candidate a denotes a resource level $C_t(a)$ for the next control interval. Service is evaluated against a service-level objective (SLO). We distinguish three cost objects: the branch charge $c_t^{\text{score}}(s_t, a)$ used to form the post-action value state, a conservative commitment $\bar{c}_t(s_t, a)$ used by the pre-actuation mask, and the measured ledger increment ΔC_t^{led} observed after actuation. The feasible set is

$$\mathcal{A}_t(b_t, s_t) = \{a \in \mathcal{A} \mid \bar{c}_t(s_t, a) \leq b_t\}. \quad (1)$$

After execution, the environment returns service reward r_t and applies

$$C_{t+1}^{\text{led}} = C_t^{\text{led}} + \Delta C_t^{\text{led}}, \quad b_{t+1} = B - C_{t+1}^{\text{led}}, \quad h_{t+1} = h_t - 1. \quad (2)$$

The objective is the finite-horizon discounted return

$$J = \mathbb{E} \left[\sum_{t=0}^{H-1} \gamma^t r_t \right], \quad (3)$$

subject to hard action affordability. In the trace environment, every candidate satisfies $c_t^{\text{score}}(s_t, a) = \bar{c}_t(s_t, a) = c(a)$. After execution, $\Delta C_t^{\text{led}} = c(a_t^{\text{exec}})$, so the update reduces to $b_{t+1} = b_t - c(a_t^{\text{exec}})$. In Kubernetes these three objects separate expected interval cost, a state-dependent safety reserve, and sampled Ready-Pod-time consumption. In both realizations, s_t, b_t, h_t form one 14-dimensional observation o_t .

We split the transition into an action-controlled component and an exogenous component. Let g encode the controlled queue, capacity, service, and action-cost effects, and let x_{t+1} denote future demand. Budget and horizon are tracked separately. Then

$$s_{t+1} = F(s_t, a_t, x_{t+1}) = f_{\text{exo}}(g(s_t, a_t), x_{t+1}). \quad (4)$$

The scheduling software already implements g : the resource target and action price are explicit, and an equation or calibration describes how capacity drains a queue. This structure provides a useful division of labor. Explicit branches compute the controlled effects, while learning estimates exogenous demand and the continuation value of the resulting state.

The trace constraint is pathwise rather than an expected-cost objective. With nonnegative exact token costs, $b_0 = B$, and $a_t \in \mathcal{A}_t(b_t, s_t)$, induction gives $\sum_t c(a_t) \leq B$. Value error can change which feasible action is chosen, but it cannot authorize an unaffordable trace action. The Kubernetes realization retains the hard pre-actuation mask while replacing exact tokens with a state-dependent commitment and a sampled Ready-time ledger; its conditional bound and empirical compliance evidence are specified in Section 4.

2.2 Why Finite Budgets Require Direct Action Comparison

The relative value of the choices in Figure 1 depends on remaining budget and horizon as well as current service gain. DAP makes this comparison explicit by constructing each candidate's queue, service, and post-cost budget before evaluating its continuation.

A shared causal forecast holds the assumed future workload fixed across candidate branches. Together with explicit action effects, it gives each score a concrete interpretation: immediate service plus the opportunity remaining after this affordable action. The fields in Table 1 supply this comparison.

Table 1. Causal observation contract used by the trace environments and the Kubernetes controller. Every transform is fitted on training data only.

Group	Fields and decision role
Demand	Current load, recent load, and burst intensity summarize the observed arrival process and drive the shared next-load forecast.
Queue/service	Queue, queue growth, active capacity, hot-resource ratio, utilization, mean/tail latency, SLO history, and resource pressure expose the known effect that an action can change.
Opportunity	Remaining-budget and remaining-horizon ratios condition the value of preserving resource for future decisions.

2.3 Action-Comparison Contract and Execution Fidelity

These are requirements on the deployed decision path. They complement the optimization objective by making affordability, action comparison, and the actuator boundary observable in the same cycle.

Definition 1 (Action-comparable execution contract). For one causal observation o_t containing normalized budget and horizon, let b_t also denote the corresponding raw, auditable external-ledger value. Given common descriptor $\hat{x}_{t+1}$, the decision record is

$$\begin{aligned}
 \mathcal{D}_t = \{ & o_t, b_t, \hat{x}_{t+1}, (\bar{c}_t(s_t, a))_{a \in \mathcal{A}}, \\
 & (\hat{o}'_{t,a}, c_t^{\text{score}}(s_t, a), r_t(a))_{a \in \mathcal{A}}, \\
 & (V_\theta(\hat{o}'_{t,a}), Q_{\text{DAP}}(o_t, a))_{a \in \mathcal{A}_t(b_t, s_t)}, (Q_{\text{DAP}}(o_t, a) = -\infty)_{a \notin \mathcal{A}_t(b_t, s_t)}, \\
 & a_t^{\text{plan}}, a_t^{\text{exec}}, \Delta C_t^{\text{led}}, C_{t+1}^{\text{led}}, b_{t+1} \}.
 \end{aligned} \tag{5}$$

A controller satisfies the contract when (C1) every candidate uses the same timestamped exogenous descriptor; (C2) its explicit action effect and exact or calibrated scoring charge construct the post-action state before continuation value is evaluated; (C3) an external ledger applies $\bar{c}_t(s_t, a)$ and removes unaffordable candidates before $a_t^{\text{plan}} = \arg \max_{a \in \mathcal{A}_t(b_t, s_t)} Q_{\text{DAP}}(o_t, a)$; and (C4) the delivered command, observed resource use, and next ledger state are recorded so that $a_t^{\text{exec}} = a_t^{\text{plan}}$ can be checked.

Four design requirements follow: (R1) shared causal context; (R2) explicit action effects; (R3) continuation at the post-cost budget and horizon; and (R4) preservation of action identity through delivery and accounting. Figure 2 maps them to planner modules. Step-keyed state, action, and event records expose the resulting invariants for replay.

These requirements separate three properties that are often conflated. *Feasibility* asks whether a decision respects the resource ledger. *Productivity* asks how much service the consumed resource produces. *Execution fidelity* asks whether the deployed action preserves the candidate ordering intended by the value computation:

$$F_{\text{exec}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1} \left[a_t^{\text{exec}} = \arg \max_{a \in \mathcal{A}_t(b_t, s_t)} Q_{\text{DAP}}(o_t, a) \right]. \tag{6}$$

Execution fidelity measures the preservation of a planned decision through delivery. Service completion, SLO outcomes, and resource use evaluate the quality of that decision; ledger closure and

Table 2. Native decision interfaces of common controller abstractions. “External” marks a field outside the method’s native decision object; DAP emits candidate consequences, affordability, scores, and actuation identity as one runtime record.

Abstraction	Candidate comparison	Budget semantics	Decision artifact	Actuator boundary
Learned policy controller	Policy logits or learned action values; explicit effects not intrinsic	External mask or learned constraint	Policy output	Integration layer
Lagrangian RL	Cost enters reward or dual update	Expected or penalized cost; hard mask external	Policy output plus multiplier	Integration layer
PDS/ADP	Exact controlled afterstate; exogenous effect in expectation	External constraint interface	Afterstate values	Selection and actuation may be external
MPC	Explicit online sequence tree under a model	Constraints inside optimizer	Plan can expose action or sequence scores	Deployment records identity of first command
Actor distillation	Planner supplies offline labels	Runtime mask external	Actor output can differ from teacher	Actor command
DAP	Shared forecast plus structured action branches	External feasible set before actuation	Complete feasible score vector	Logged feasible argmax delivered directly

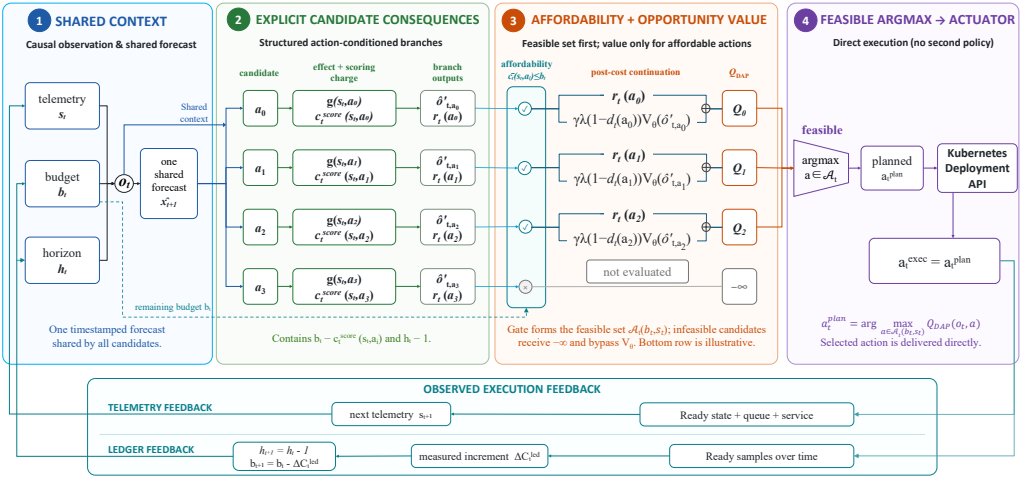


Fig. 2. The four-module DAP action-comparison contract. One observation and shared forecast feed explicit candidate consequences. External commitments form the feasible set before learned continuation-value evaluation; the maximizer reaches the Kubernetes Deployment API directly, and separate telemetry and ledger feedback close the loop.

feasibility evaluate accounting. Keeping these quantities distinct lets an evaluator locate a discrepancy in action valuation, selection, actuation, or resource measurement. The native interfaces in Table 2 motivate bringing these relations together in one runtime object.

3 Direct Action Planning

DAP combines learned exogenous demand and continuation value with explicit resource-transition equations. Figure 2 shows how shared context, candidate branches, post-cost scoring, and direct actuation implement the four requirements. Measured resource feedback closes the decision loop.

3.1 Causal Context Shared across Actions

Module 1 in Figure 2 combines causal telemetry s_t , budget b_t , and horizon h_t into $o_t = \phi(s_t, b_t, h_t)$. Here ϕ constructs the 14 features in Table 1 using training-fitted normalization. Current reward uses observed load x_t ; the forecaster, with parameters ψ , predicts next load once,

$$\hat{x}_{t+1} = f_\psi(o_t), \quad (7)$$

and the same timestamped value is supplied to every candidate action. The action-invariant forecaster receives all 14 causal features and uses a $14 \rightarrow 32 \rightarrow 32 \rightarrow 1$ Tanh network with non-negative output. Training minimizes $\mathcal{L}_{\text{load}} + 4\mathcal{L}_Q + 2\mathcal{L}_{\text{rank}}$: Smooth-L1 log-load error, normalized feasible-branch Q error, and pairwise ranking loss with margin 0.02. Validation selects the epoch by action error plus 0.25 normalized Q error and 0.05 load error. The value is fitted first and frozen during 24 forecast-training epochs. For training state j , targets $Q_j^{\text{target}}(a) = r_j(a) + \gamma(1 - d_j)V_\theta(o'_{j,a})$ use observed next-load branches; predictions substitute the common forecast. The recorded terminal bit is d_j . $\mathcal{L}_Q$ is Smooth-L1 error after dividing both Q vectors by the training feasible-Q standard deviation (floored at one). $\mathcal{L}_{\text{rank}}$ is the pairwise hinge loss on normalized Q differences, with the target ordering supplying the sign. Adam uses learning rate 10^{-3} , batches of 256, and gradient clipping at 5.

The shared forecast implements R1; its recorded timestamp supports independent checks of causal context across all branches.

3.2 Explicit Action-Conditioned Transitions

The trace action is a one-step capacity level, not a persistent increment: $C_t(a) = C_{\text{base}} + \delta(a)$, with token costs (0, 1, 2, 4) and nominal extras (0, 1, 2.2, 4.8). The effect lasts for the current trace step and the next decision chooses a fresh level. Dataset-specific capacity calibration is fitted on training data and shared by all methods. In the Kubernetes instantiation the same ordered action identifiers map instead to absolute Deployment targets (1, 2, 3, 5), which persist until the next command; its no_op identifier denotes the base/recovery target 1, so it requests scale-down when the current target is larger. For current arrival x_t , queue q_t , and candidate capacity $C_t(a)$, each copied branch applies

$$u_t = q_t + x_t, \quad y_t(a) = \min(u_t, C_t(a)), \quad (8)$$

$$q_{t+1}(a) = \text{clip}(u_t - y_t(a), 0, q_{\text{max}}), \quad (9)$$

followed by per-step clearance, a queue-derived latency proxy, and its SLO indicator:

$$\rho_t(a) = \frac{y_t(a)}{\max(u_t, 1)}, \quad (10)$$

$$\ell_t^{\text{tail}}(a) = 1 + \frac{2q_{t+1}(a)}{C_t(a)}, \quad (11)$$

$$z_t(a) = \mathbf{1}[\ell_t^{\text{tail}}(a) > \ell_{\text{SLO}}]. \quad (12)$$

The common simulator reward is

$$r_t(a) = \rho_t(a) - 0.04q_{t+1}(a) - 0.08\ell_t^{\text{tail}}(a) - 2z_t(a). \quad (13)$$

After inserting the shared forecast, the branch emits

$$\hat{o}'_{t,a} = \phi(s'_{t,a}(\hat{x}_{t+1}), b_t - c_t^{\text{score}}(s_t, a), h_t - 1). \quad (14)$$

Here $s'_{t,a}(x) = f_{\text{exo}}(g(s_t, a), x)$. The hat marks predicted load; training branches $o'_{j,a}$ use observed transitions. Figure 2 labels candidates $a_0, \dots, a_3$: g supplies their effects, and ϕ inserts the updated budget and horizon.

3.3 Affordability and Continuation Value

A $14 \rightarrow 64 \rightarrow 64 \rightarrow 1$ Tanh network learns the continuation value $V_\theta(o)$, with parameters θ . For fitting state j , fitted value iteration (FVI) regresses the target

$$Y_j = \max_{a \in \mathcal{A}_j(b_j, s_j)} [r_j(a) + \gamma(1 - d_j)V_{\bar{\theta}}(o'_{j,a})], \quad (15)$$

where $\bar{\theta}$ is the frozen preceding-round parameter vector. Training uses 18 rounds indexed 0–17. At deployment, each affordable candidate receives

$$Q_{\text{DAP}}(o_t, a) = r_t(a) + \gamma\lambda(1 - d_t(a))V_\theta(\hat{o}'_{t,a}), \quad d_t(a) = \mathbf{1}[h_t = 1]. \quad (16)$$

Here λ weights continuation, and $d_t(a)$ suppresses it at the final decision. Figure 2 abbreviates $Q_{\text{DAP}}(o_t, a_i)$ as Q_i . Its gate uses $\bar{c}_t$, whereas c_t^{score} sets the branch's post-cost budget. These charges coincide for trace tokens; Kubernetes distinguishes them. Unaffordable actions receive $-\infty$ and bypass V_θ ; the masked a_3 row illustrates a state-dependent outcome. Both deployed planners bypass V_θ at $h_t = 1$; terminal-mask tests verify that immediate rewards determine the final scores.

Training has three non-overlapping chronological roles. The fitting block trains forecast and value candidates; the selection block chooses a retained value round and $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$; the development-evaluation block supports information-matched controls. The final temporal evaluator loads a frozen checkpoint and performs inference only. Selection retains the immediate planner ($\lambda = 0$) as a candidate and applies fixed validation guardrails: completion within 0.005, SLO violation within 0.01, and added cost no greater than $0.10B$ relative to that reference. Eligible candidates are ordered by return, completion, SLO, cost, and deterministic tie breaks. These rules freeze the operating point before evaluation.

For each fitting state, data collection copies the environment and executes every affordable action, storing its reward, next observation, feasibility, and terminal bit; 24 training and six validation episodes are collected per domain. Each FVI round freezes the preceding network as its target, computes the four masked Bellman branches, and fits the state maximum with Smooth-L1 loss using Adam (learning rate 10^{-3}), batches of 256, two epochs per round, and gradient norm 5. A DAP fitting state expands every affordable branch using the known transition equations; these branch evaluations are model computations, not additional environment interactions. The network width is 64, output scaling is fitted from training-reward RMS, and checkpoints $\{0, 2, 4, 6, 8, 10, 12, 14, 16, 17\}$ are the ten prespecified rounds retained for validation selection. Section 5 specifies the shared interaction and safety protocol for comparisons with learned policies.

3.4 Feasible Selection and Direct Actuation

The controller executes the deterministically tie-broken feasible maximizer,

$$a_t^{\text{plan}} = \arg \max_{a \in \mathcal{A}_t(b_t, s_t)} Q_{\text{DAP}}(o_t, a), \quad (17)$$

and issues it directly, so $a_t^{\text{exec}} = a_t^{\text{plan}}$, preserving the selected candidate as the delivered command. The cycle record in Definition 1 includes step-keyed state, action, and event entries with raw and normalized observations, field timestamps, missing-data policies, branch charges, feasible bits, API latency, and sampled ledger intervals. Replay checks shared context, post-cost states, feasible argmax, target mapping, delivery identity, selected-action feasibility, and ledger closure. Figure 2

shows separate feedback for telemetry s_{t+1} and measured cost ΔC_t^{led} . Its update $b_{t+1} = b_t - \Delta C_t^{\text{led}}$ is equivalent to the cumulative ledger equation in Section 2.

For $|\mathcal{A}| = 4$, online work is one forecast plus $O(|\mathcal{A}|)$ one-step transition and value calls. A depth- K , width- W beam MPC performs $O(KW|\mathcal{A}|)$ branch expansions. DAP amortizes longer-horizon reasoning into V_θ while preserving explicit one-step action semantics and exposing an action-specific consequence trace. Section 4 maps this module to a Kubernetes Deployment actuator and an independently measured Ready-Pod-time ledger.

Relation to post-decision value. PDS/ADP separates the controlled afterstate from subsequent exogenous uncertainty [8, 25]. DAP carries this decomposition through deployment by retaining candidate effects, affordability, scores, and actuator identity in one record. This makes the comparison executable and replayable. Section 5.5 tests a matched PDS/ADP representation, and Table 2 identifies the exposed runtime artifacts.

4 Kubernetes Realization

The Kubernetes realization maps the contract to three concrete interfaces: telemetry supplies causal state, the Deployment API receives absolute resource targets, and an independent Ready-time ledger measures consumption. We implement DAP as an external Python controller for a local cluster. Each 5-second cycle reads Deployment status, Pod readiness, per-Pod Prometheus counters, queue depth, request rate, latency, CPU, SLO, and remaining budget. The planner maps its four actions to absolute replica targets $\{1, 2, 3, 5\}$ and issues a Deployment scale request through `kubect1`. A target remains in force until the next cycle; the “no-op” identifier is explicitly the base/recovery action, not a hold-current command. Thus a base action issued from target 5 requests scale-down. The controller does not modify the application or Kubernetes scheduler.

The budget charges Ready replicas above one base replica. A separate tracker samples readiness approximately every 0.25 seconds; the pre-actuation reserve admits candidates that fit the remaining sampled ledger. After each run, the controller returns to the base target and continues accounting until scale-down is observed. Kubernetes readiness probes ensure that a new Pod is charged and used only once it becomes Ready; Pending or non-Ready Pods are excluded from Service traffic [45], while the target-based reserve still accounts for their possibility of becoming Ready during the next interval. The controller uses the Deployment scale interface exposed to the native Horizontal Pod Autoscaler (HPA) [44], preceded by DAP’s planner and Ready-time ledger. It follows the controller-oriented design of Borg and Kubernetes [9, 48].

Let $R(\tau)$ be the Ready-replica process and let one replica be the uncharged base capacity. The target continuous-time cost is

$$C_{\text{ready}}(t) = \int_0^t \max(R(\tau) - 1, 0) d\tau. \quad (18)$$

For observations (τ_k, R_k) , the implementation records the left-step approximation

$$\hat{C}_k = \sum_{j=1}^k \max(R_{j-1} - 1, 0)(\tau_j - \tau_{j-1}). \quad (19)$$

Every sample interval and Ready count is logged, so $\hat{C}_k$ is independently recomputable; requested replicas are logged separately. For control period Δ , current Ready count R_t , candidate absolute target $n(a)$, and calibrated lifecycle guard G , the mask uses

$$\bar{c}_t(R_t, a) = (n(a) - 1)_+ \Delta + \max\{(R_t - 1)_+, (n(a) - 1)_+\}G. \quad (20)$$

It admits a non-base target only when $\hat{C}_t + \bar{c}_t(R_t, a) \leq B$; the base target remains the recovery command. The reserve is conditional on a bounded control interval and a Ready tail covered by G , including outstanding replica commitments. Under these conditions, an admitted action plus return to base fits the remaining ledger. The implementation measures the left-step Ready integral, so the live budget endpoint is sampled-ledger compliance. Exact token affordability applies to the trace environment; continuous-time accounting additionally requires independently bounded command delays, Ready tails, and pending commitments.

The guard is calibrated from scale-up and return-to-base observations. For each workload profile and target, the calibration takes the maximum observed Ready-withdrawal delay, adds its sampling gap and a one-second margin, and validates the result on separate repetitions. Section 5.6 reports these measurements and the guards used in live comparisons.

This architecture separates three concerns: live metric collection constructs the 14-field observation with provenance; the planner ranks feasible actions; and the actuator plus budget ledger enforce deployment and resource semantics. The runtime uses one timestamped snapshot per cycle. Missing Pod metrics are retried three times, then recorded as missing and omitted from service features if unavailable; the independent current Ready channel is read from Deployment status and remains separate from stale learned features in the lag fault experiment. If the Deployment query itself fails, the run is marked incomplete and the cleanup path requests the base target. Consequently, service noise can affect ranking, while the Ready-Pod ledger and pre-actuation mask remain separate from the learned score.

The live implementation preserves the learned transition semantics: the value branch receives interval-average Ready capacity, while an independent current-Ready channel supplies the hard affordability mask. Utilization and exponentially weighted moving-average (EWMA) features use the same causal windows as training. To keep burst estimates causal and action-comparable, the runtime computes one envelope for all actions:

$$\hat{x}_{t+1} = 1.10 \max(\hat{x}_{t+1}^{\text{neural}}, x_t, \hat{x}_{t+1}^{\text{EWMA}}).$$

The contract transfers across trace and Kubernetes instantiations with dataset- and deployment-specific fitted parameters. Trace checkpoints are fitted on each workload's training split and frozen before temporal testing. Kubernetes uses separate checkpoints and an independently measured Ready-capacity/startup/scale-down calibration, selects controller settings on chronological validation inputs, and freezes both before live test plans. The GenTD26 continuation coefficient is selected on disjoint validation plans within the same state, action, branch, mask, and actuator interfaces.

Component ownership follows the same interfaces. The workload driver owns a frozen request plan and client-side outcomes; the worker service owns admission, bounded queueing, CPU work, timeout accounting, and Prometheus metrics; Kubernetes owns Pod lifecycle and Service routing; and the controller owns state construction, planning, budget accounting, and replica patches. Only one controller is active in a run. A planner, checkpoint, or API failure triggers the base-replica cleanup path and prevents the run from being marked complete. This separation attributes planning, actuation, and request-accounting outcomes to the responsible component.

Observation records retain raw and normalized values, timestamps, freshness, and missing-data policies. Tests cover transition consistency, causal inputs, terminal values, absolute targets, stale safety data, and delivered argmax before formal runs are admitted to analysis.

5 Evaluation

The evaluation examines the contract from three complementary perspectives: service productivity, the mechanism of action comparison, and deployed execution. These perspectives correspond to the following research questions.

RQ1 – Effectiveness: How does DAP translate a fixed resource budget into service quality compared with learned scheduling policies?

RQ2 – Mechanism: How do explicit action effects and continuation value contribute to action ordering and planning efficiency?

RQ3 – Deployment: How does the contract support independent verification and effective resource control in a deployed system?

5.1 Workloads and Experimental Design

Azure Functions 2019 provides per-minute invocations in HTTP and asynchronous domains [32, 39]; days 1–8, 9–11, and 12–14 form train, validation, and test. GenTD26 is drawn from the Alibaba clusterdata trace and aggregated into ten-minute serving-demand bins [28]; its split is 60%/20%/20%. Each validation segment is further partitioned chronologically into 40% model validation, 30% checkpoint selection, and 30% development evaluation. Clipping, normalization, and capacity calibration use training data only. Each dataset fits its own controller, while the shared object is the four-module contract. Temporal slices are frozen inference re-evaluations of previously accessed project data; code, checkpoint, and window hashes bind their provenance.

The horizon is 64, $\gamma = 0.99$, and budgets are {48, 72, 96, 120, 144}. The main matrix contains two datasets, five budgets, and ten training seeds (820–829) with five evaluation episodes per domain. Mask-binding frequency measures how often scarcity restricts the candidate set, alongside the zero-overspend invariant. GenTD26’s domains are text-to-image and image-conditioned serving. Every method records zero token overspend under the common mask.

5.2 Comparators and Statistical Analysis

External comparators include Double DQN, PPO, PPO-Lagrangian, CPO, P3O, Budgeted Fitted-Q, A2C, PID-Lagrangian, and a disclosed port of LCPO [1, 10, 23, 37, 40, 46, 52]. The baselines span model-free, constrained, budget-conditioned, and context-aware learning. Each receives 102,400 environment interactions and a 64-unit network width. Immediate planning, MPC-2/4/8, black-box transitions, actor distillation, budget/horizon removal, and PDS/ADP are information-matched controls. All methods share observations, interactions, actions, rewards, trace windows, normalization, and the affordability mask; native policy and dual losses remain unchanged. This common safety-admission protocol measures the service productivity of choices within the same feasible action set. DAP and MPC expose computable effects, while model-free methods infer them. Frozen manifests record optimizer settings, update counts, search ranges, branch counts, and wall-clock summaries.

The trained seed is the trace unit. We use 20,000 paired seed-block bootstrap resamples for 95% intervals, paired Cohen’s d_z , wins/ties/losses, two-sided Wilcoxon tests, and Benjamini–Hochberg correction [2, 6, 7]; cost matching stays within observed budget frontiers. Kubernetes uses paired request plans, paired-bootstrap intervals, exact sign-flip tests, and the registered Holm family, retaining all seeds and failures. Matched trace, learned-transition, and same-initialization feature controls use exact sign flips and Holm families of 24, 16, and eight endpoints; the matched live family contains 18 endpoints. Budgets are averaged within each of ten paired seed or plan blocks. The matched studies enumerate all 2^{10} sign assignments and use bootstrap seed 20261002; intervals are pointwise and tests carry familywise correction.

5.3 Measures and Reproducibility

Return is the optimization objective. Episode completion is $\sum_t y_t / \max(\sum_t x_t, 1)$, whereas ρ_t is the per-step clearance term used only in reward. With initial queue zero, overflow $w_t = (q_t + x_t - y_t - q_{t+1})_+$ makes the conservation identity $\sum_t x_t = \sum_t y_t + \sum_t w_t + q_H$ explicit; unfinished and dropped work are not completions. Trace latency and SLO are queue-model proxies, not measured request percentiles. Kubernetes completion counts successful HTTP requests over all scheduled requests; latency percentiles condition on successful responses. The matched live study also reports the fraction of all requests that fail or exceed the SLO, counting fast failures as violations. Trace tables average episodes equally, assigning zero completion to zero-demand episodes (0/500 Azure and 20/500 GenTD26 episodes per method). A descriptive request-weighted check gives DAP completion 96.82%/79.56%; GenTD26 additionally has 5.15% dropped and 15.30% unfinished arriving mass. The 5,000-episode audit closes mass conservation to 1.31×10^{-12} and finds no token overspend. These aggregation conventions leave the primary comparisons unchanged. Resource use is total action cost in the trace environment and additional Ready-replica-seconds in Kubernetes. We report budget utilization, overspend, Pareto cell counts, matched-frontier effects, action disagreement, Q-star regret, execution fidelity F_{exec} , and decision latency within their corresponding evaluation families.

Frozen inference manifests bind dataset slices, configurations, checkpoints, code, and outputs. The main trace matrix comprises 100 cells; PDS covers 100 development and 100 temporal cells. Kubernetes comparison and robustness studies contain 260 runs; matched controls contain 160 runs and 5,120 action records. All specified seeds and outcomes are retained with the paired statistics defined above. Repeated steps and requests remain observations within their seed or plan block.

AI-assistance disclosure. The authors formulated the research questions and DAP design, planned the experiments, and interpreted the results. Under their instructions, an AI coding assistant supported controller and adapter implementation, testing, translation of experimental protocols into runnable scripts, trace/GPU/Kubernetes execution, statistical and exact-DP computations from recorded outputs, and visualization. Code, protocols, checkpoints, and measurements retain the provenance of this assistance. The authors are responsible for validating these artifacts and the final conclusions.

5.4 RQ1: Service–Resource Effectiveness

DAP produces favorable service–resource operating points on both workloads, with the form of the gain depending on workload demand (Table 3 and Figure 3). On GenTD26, DAP has the best mean return, completion, SLO violation, and cost among the compared methods. Relative to Double DQN, it gains 45.764 return and 6.61 completion points while reducing SLO violation by 5.27 points and cost by 36.186 units. Relative to CPO, the corresponding improvements are 57.031 return, 7.29 completion points, 6.22 SLO points, and 40.346 cost units. Every trained-seed block favors DAP, and all four corrected comparisons have $q = 0.001953$.

Azure exposes two operator-relevant regimes. Against CPO, DAP improves return by 3.762 while reducing SLO violation by 1.62 points and cost by 12.144 units. At supported cost-matched points, DAP exceeds the interpolated CPO return frontier by 12.046 and reduces SLO violation by 5.98 points in 45/50 cells. DAP and the cost-lean Double DQN point form distinct Pareto choices: DAP provides higher return and lower SLO violation when additional budget is available. Against the separately ported LCPO mechanism, all four paired outcomes favor DAP in every trained-seed block on both datasets ($q = 0.001953$).

Table 3. Mean application operating points over five budgets and ten trained seeds. Arrows show direction; bold marks the best value in each dataset and column, and † marks a Pareto-nondominated row over all four outcomes.

Dataset	Method	Return ↑	Completion ↑	SLO ↓	Cost ↓
Azure	DAP †	23.783	.9711	.1038	73.8
Azure	Double DQN†	14.295	.9736	.1186	62.9
Azure	PPO	6.715	.9574	.1967	91.3
Azure	A2C	-8.861	.9374	.2887	92.9
Azure	PPO-Lagrangian	6.465	.9561	.2068	92.5
Azure	CPO	20.021	.9684	.1200	86.0
Azure	P3O	.431	.9551	.2251	90.3
Azure	PID-Lagrangian	8.033	.9578	.1977	92.2
Azure	Budgeted Fitted-Q	-36.729	.9183	.4341	95.9
Azure	LCPO	7.833	.9523	.2044	93.8
GenTD26	DAP †	-105.195	.8851	.1479	39.0
GenTD26	Double DQN	-150.959	.8191	.2006	75.2
GenTD26	PPO	-215.266	.6979	.2691	95.6
GenTD26	A2C	-208.719	.7174	.2715	89.4
GenTD26	PPO-Lagrangian	-222.110	.6903	.2716	96.0
GenTD26	CPO	-162.225	.8122	.2101	79.4
GenTD26	P3O	-221.485	.6905	.2711	96.0
GenTD26	PID-Lagrangian	-219.204	.6930	.2706	96.0
GenTD26	Budgeted Fitted-Q	-199.016	.7322	.2604	93.0
GenTD26	LCPO	-170.505	.7913	.2268	83.0

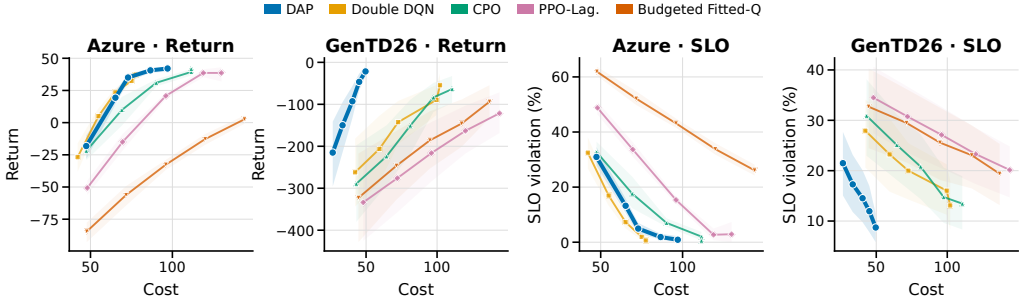


Fig. 3. Public-workload service-cost frontiers. The first two panels show return (higher is favorable) and the last two show SLO violation (lower is favorable), each against observed cost for Azure and GenTD26. Each point is one registered budget averaged over ten independently trained seeds; translucent ribbons are seed-bootstrap 95% intervals for the vertical endpoint, and lines connect observed budgets only.

Figure 4 reports the corresponding paired effects after averaging budgets within each trained seed.

These results connect finite-budget planning to an operator's choice of service-resource regime. On GenTD26, the DAP mean operating point improves all four outcomes relative to every comparator. On Azure, DAP combines a favorable return-SLO frontier with lower cost than CPO, while

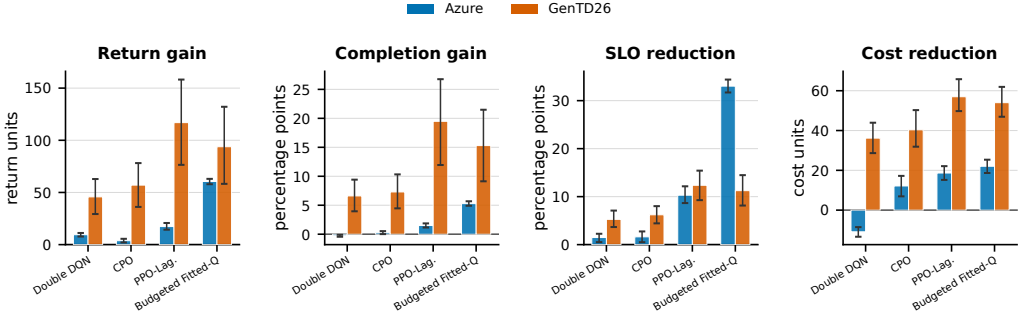


Fig. 4. **DAP improvement over baseline.** Define $\Delta R = R_{\text{DAP}} - R_{\text{baseline}}$ and $\Delta C = C_{\text{DAP}} - C_{\text{baseline}}$; SLO reduction is $\text{SLO}_{\text{baseline}} - \text{SLO}_{\text{DAP}}$, and cost reduction is $\text{Cost}_{\text{baseline}} - \text{Cost}_{\text{DAP}} = -\Delta C$. The four endpoints are therefore oriented so that positive bars favor DAP. Whiskers are paired-bootstrap 95% CIs over ten trained seed blocks after averaging five budgets; completion and SLO use percentage points, while return and cost retain native units.

the DAP and Double DQN points offer different allocations of service and resource. The frontiers make this choice explicit across the evaluated budgets.

5.5 RQ2: Mechanisms of Action Comparison

The mechanism study separates the representation of action effects from the estimation of future value. Matched controls test these components on workload traces, and an exhaustive dynamic-programming diagnostic directly measures the resulting action order.

Explicit effects and continuation. Matched temporal controls cover all 100 dataset–budget–seed cells, with the original checkpoints and matched windows. Table 4 shows that Azure DAP improves return, completion, and proxy SLO over immediate planning and MPC-8 after correction. Against a learned full transition with the same frozen continuation weight, both datasets improve on all four endpoints. DAP’s mean decision times are 0.81/0.95 ms, versus 144.88/189.04 ms for MPC-8 in this run.

The budget/horizon comparison pairs a full-feature value model with a masked model trained from the same initialization, data, batches, GPU, and selected round. This pairing isolates the information available to the value model. Table 4 reports this paired control alongside the frozen-checkpoint comparisons. The learned transition has width 96 and trains for 24 epochs; feature controls retain 18 FVI rounds, two epochs per round, and the frozen continuation weight. Training uses PyTorch 2.11 on an RTX 5090; planning runs on CPU. Frozen selection chooses zero continuation in 8/50 Azure and 16/50 GenTD26 checkpoints; those cells reduce to immediate scoring.

Action-ordering accuracy. Exact DP supplies an exhaustive reference for the branch comparison by evaluating every feasible state–budget–horizon–action tuple.

For state s , budget b , horizon h , and candidate a , backward DP defines $Q^*(s, b, h, a)$ and regret $Q^*(s, b, h, a^*) - Q^*(s, b, h, a)$, with the lowest-index exact maximizer a^* . True-transition, black-box, and structured branches reach .9553/.8805/.9460 agreement and .0191/.0613/.0188 regret; the structured branch reduces Q-star regret 69.33% and large-regret errors 99.49%. Table 5 reports the exhaustive ordering diagnostic.

The diagnostic fixes horizon 16, arrivals $\{1, 2, 4\}$, queues 0–6, budgets 0–12, costs $(0, 1, 2, 3)$, capacities $(1, 2, 3, 4)$, and $\gamma = .99$, with stable, burst, and periodic load scenarios. Backward DP

Table 4. Temporal mechanism controls over all five budgets and ten paired training seeds per dataset. Differences are DAP minus comparator; † instead compares the full-feature retrain with its identically initialized masked-feature counterpart. Return includes pointwise 95% intervals; completion and SLO differences use percentage points. Stars indicate Holm-adjusted $p < .05$ within each fixed study family (24 trace, 16 learned-model, eight matched-feature endpoints).

Dataset	Comparator	Δ return [95% CI]	Δ comp.	Δ SLO	Δ cost
Azure	Immediate	+8.72[+7.32, +10.20]*	+1.11*	-4.52*	-3.46
	PDS/ADP	+0.73[-0.24, +1.62]	+0.04	-0.24	-0.80
	MPC-8	+4.80[+3.98, +5.68]*	+0.69*	-2.39*	-3.46
	Learned trans.	+5.38[+3.81, +7.01]*	+0.65*	-2.54*	-7.73*
	No B/H†	+4.68[+1.72, +8.10]	+0.32	-2.07	-0.65
GenTD26	Immediate	+1.79[-0.13, +4.25]	+0.20	-0.04	-0.24
	PDS/ADP	+0.47[-0.58, +1.65]	+0.07	+0.04	-0.00
	MPC-8	-0.82[-2.51, +0.85]	+0.08	+0.22	-0.07
	Learned trans.	+19.30[+11.48, +27.98]*	+3.79*	-2.07*	-19.09*
	No B/H†	-1.91[-9.11, +7.65]	-0.65	+0.25	+7.26

Table 5. Exhaustive exact-DP action-ordering diagnostic over every feasible state-budget-horizon-action tuple. Lower Q-star regret is favorable.

Planner branch	Action agreement	Q-star regret
True transition + learned value	.9553	.0191
Black-box learned transition	.8805	.0613
Structured action effects	.9460	.0188

evaluates the full feasible state-budget-horizon-action grid over three scenarios and five seeds; ties use the lowest action index.

The large-regret set is the upper quartile of positive Q-star regret among black-box-model disagreements: threshold 0.7073, 1,365 states, and 1,365 versus seven errors for black-box and structured branches. The 69.33% and 99.49% are exhaustive-grid ratios, while bootstrap uncertainty is reported for 1,800 rollouts. This grid quantifies the ordering mechanism; workload-scale behavior is reported in the trace and live evaluations, with independent paired Kubernetes runs.

Alternative post-decision value representation. The preregistered PDS comparator tests a closely related value representation. It applies the exact action, reward, queue, budget, and horizon transition, then removes next-load fields before the exogenous value is revealed. The afterstate network scores feasible actions as immediate reward plus discounted post-decision value. It shares the frozen DAP backbones, data, widths, candidate rounds, continuation weights, guardrails, budgets, and seeds; all development and temporal cells use the same code and contract hashes and complete with zero overspend.

The methods select different actions in 3.69–13.47% of Azure states and 1.09–4.11% of GenTD26 states. Their complete temporal contrasts appear in Table 4. The PDS adapter also provides a portability test: the same execution interface carries a different value representation while preserving its clean scores and actions. This separates the reusable verification capability from the choice of continuation model.

The matched-transition and exact-DP results support the same interpretation: exposing known action effects improves the ordering on which the controller acts. Azure's continuation gains show how the post-cost value can further improve service beyond immediate scoring. DAP obtains these gains with sub-millisecond mean planning in the matched study, while the PDS adapter demonstrates that the verification interface also accommodates an alternative value representation.

5.6 RQ3: Verification and Deployed Resource Control

RQ3 connects independent checks across DAP and PDS with live resource control. Paired Kubernetes plans measure service outcomes, delivered commands, and resource accounting through a real orchestration API.

Independent verification across representations. The contract makes relationships between software components executable. To test this capability, a trace adapter injects faults before branch construction, selection, mapping, delivery, or ledger updates take effect. The seven fault types are stale cycle context, inconsistent forecast, incorrect post-cost budget, nonmaximal selection, target remapping, delivery override, and omitted ledger charge. The checker receives independently witnessed context and actuator receipts without fault labels. For example, a recorded decision selects capacity 6 with score 1.801; a delivery override executes the valid, affordable capacity 5 with score 1.785. Local validity checks accept either target, while the independent receipt exposes the change in planner-to-actuator identity. All 640 clean DAP/PDS decision contexts preserve the original scores and actions. On independently witnessed activated faults, the cross-boundary checker detects and localizes 1,940/1,940 DAP and 1,614/1,614 PDS violations; local validity, bounds, and affordability assertions detect 6/3,554, while passive logs perform no automatic detection. Clean false positives are 0/640; median checker-only time is 7.80 microseconds (95th percentile 8.69), excluding evidence collection and persistence. A separate +10% coherent-capacity control yields 50 changed actions in 640 contexts while preserving record consistency, and the checker correctly retains those internally consistent records. Each injected run executes one copied context; inactive cases and PDS forecast injections are excluded from the applicable-fault denominator. The executable specification is implemented through cross-boundary assertions, with the same checks supporting both planning representations.

Matched live design. The matched study separates value, lookahead, and reserve effects by fixing the model and execution path and pairing DAP, immediate scoring, MPC-4, and a 45-second DAP reserve at budgets 128 and 256, using ten request plans per profile (160 completed runs). Plans map 32 consecutive trace bins to 32 five-second intervals using a training-fitted rate conversion and ten fixed activity strata. Plan seeds are 2026100201–2026100210; method order rotates across seeds and reverses at the second budget, whose order also alternates. Azure/GenTD26 use frozen continuation weights 1/.05 and HTTP SLOs .25/1 seconds. MPC uses depth four and zero terminal value; the 45-second arm changes DAP's mask reserve alone. The shared kind node runs Kubernetes 1.35.1 and Python 3.11.6. Each worker requests .25 CPU/96 MiB and is limited to one CPU/256 MiB, with one-second readiness probes and a bounded queue/CPU-work HTTP service.

Service gains under matched execution. Table 6 holds the calibrated model and execution path fixed. Averaged over both budgets within each of ten paired Azure plans, DAP increases completion over immediate scoring by 6.77 points (95% CI [5.62, 7.88]) and reduces cost by 73.95 Ready-replica-seconds (CI [-85.51, -62.96]), a 54.4% saving. Against matched MPC-4, cost falls by 71.15 (CI [-85.86, -56.68]), or 53.4%. All three effects have Holm-adjusted $p = .0352$ across the fixed 18-endpoint family. A scheduling amendment resolved exact run identifiers and retained every

Table 6. Matched live controls: means over ten paired plans, averaging budgets 128 and 256 within each plan. Completion and failure-or-SLO count all scheduled requests. Cost is sampled Ready-replica-seconds; binding is the percentage of decisions with a masked candidate. Full budget-specific results and all 18 corrected contrasts appear in the replication package.

Profile	Method	Complete (%)	Fail/SLO (%)	Cost	Binding (%)
Azure	DAP	92.98	61.10	61.98	0.00
Azure	Immediate	86.22	53.27	135.93	19.69
Azure	MPC-4	88.56	47.96	133.13	14.53
Azure	DAP (45 s)	92.90	60.62	56.88	57.66
GenTD26	DAP	93.70	28.02	98.79	14.06
GenTD26	Immediate	89.05	37.24	82.34	10.62
GenTD26	MPC-4	90.11	36.51	82.07	10.78
GenTD26	DAP (45 s)	92.74	36.42	78.99	66.72

completed run; the interrupted preparation produced no requests or actions. An eight-plan sensitivity omitting the two earlier plan seeds retains the Azure completion–cost pattern (+6.79 points and -70.83 seconds versus immediate). The 45-second control exposes the reserve’s effect on the feasible set: Azure/GenTD26 binding rises from 0/14.06% to 57.66/66.72%. These comparisons show how continuation and the reserve shape different parts of a decision: the former scores remaining opportunity, while the latter determines which commands can be considered.

Execution fidelity and control cost. Across all 160 matched runs, 3,390,344 requests reconcile exactly and all 5,120 logged actions maximize their feasible score vectors with the specified target mapping. Every sampled ledger recomputes exactly, with zero budget violations or controller deadline misses. The complementary 260-run controller and robustness studies verify 3,840 DAP decisions, request closure, and restored base readiness. Their affordability mask activates in 33/120 controller runs and 248 decisions, removing 606 candidate opportunities. A replay verifier additionally localizes each of six single-field mutations in 3,200 decision records (19,200/19,200), checking forecast equality, feasible argmax, delivery identity, target mapping, affordability, and ledger closure.

The Kubernetes loop uses Prometheus telemetry, a five-second period, a 32-cycle horizon, and 128/256 Ready-replica-second ceilings. Mean inference/full-loop times in the controller study are 0.98/0.90 ms and 0.248/0.282 s for Azure/GenTD26. The corresponding trace timing benchmark measures 0.630/0.613 ms for DAP, 46.46/37.84 ms for MPC-4, and 181.24/142.23 ms for MPC-8; Section 5.5 reports timing in the matched mechanism study. The amortized value keeps online comparison small relative to the control period. Deadline checks cover every run, and the artifact records tail latency and controller resource use.

Controller operating points. In the scheduler comparison, Azure DAP raises completion by 7.97 points (95% CI [6.48, 9.52]) and reduces Ready cost by 49.53 replica-seconds (CI [-64.75, -35.18]) over MPC-4. Under binding GenTD26, completion rises 3.68 points (CI [1.40, 6.15], Holm-adjusted $p = 0.0264$), with a mean cost difference of +0.049 (CI [-11.40, 10.39]). The two profiles provide service gains at lower or nearly equal mean cost; SLO intervals accompany both.

MPC-4 and Threshold use settings selected on a disjoint 180-run screen. The native-controller study evaluates DAP, HPA, and Kubernetes Event-Driven Autoscaling (KEDA) on common frozen plans (Table 7). Both use one to five replicas and zero scale-up/down stabilization. HPA targets 65% CPU with a two-Pod/10-second scale-up policy; KEDA polls queue depth every 10 seconds, with

Table 7. Direct native-controller comparison on the tested Kubernetes cluster (means over ten paired plans). Cost is additional Ready-replica-seconds; SLO uses the latency-only violation definition. Bold marks the favorable value in each profile and column.

Profile	Method	Completion $\uparrow$	SLO $\downarrow$	Ready cost $\downarrow$
Azure	DAP	.9654	.3705	48.37
Azure	HPA	.9490	.6157	84.33
Azure	KEDA	.9520	.4116	119.58
GenTD26	DAP	.9723	.0690	227.17
GenTD26	HPA	.8265	.2002	96.46
GenTD26	KEDA	.8727	.2440	59.81

threshold four, a 20-second cooldown, and a two-Pod/5-second scale-up policy. Both allow 100% scale-down per five seconds. The native controllers reserve 45 seconds for recovery; DAP uses the calibrated 3.2768/3.2632-second guards. These rows describe the complete configured service-resource operating points.

Table 7 exposes all service-resource operating points. On Azure, DAP uses 35.96/71.21 fewer Ready-replica-seconds than HPA/KEDA, with completion .9654 and latency-SLO violation .3705. GenTD26 DAP provides a .9723-completion, .0690-SLO operating point at 227.17 Ready-replica-seconds. The four paired cost contrasts have Holm-adjusted $p = .0234$; the service means describe the configured operating points ($p = .0625-.2227$).

Lifecycle calibration and sensitivity. The deployed controller uses startup-derived guards $G = 3.2768$ seconds (Azure) and 3.2632 seconds (GenTD26). A direct lifecycle study measures 60 scale-up/return-to-base pairs across both profiles and targets 2, 3, and 5. For each profile-target combination, the first five repetitions calibrate the maximum observed Ready-withdrawal delay plus its sampling gap and a one-second margin; the next five validate that rule. The resulting guards are 3.068/3.015 seconds, and all 30 validation withdrawals fit them (maxima 1.847/1.867 seconds). Pod deletion is measured separately (maxima 2.530/2.549 seconds). These measurements use one shared node under background activity. The paired runs use the larger startup-derived guard and continue accounting through observed return to base.

DAP preserves its registered service bounds under deterministic 10% sample-and-hold dropout on both profiles, while the independent Ready channel remains intact. Across dropout, one-cycle observation lag, and five-second readiness delay, all 60 fault runs preserve deadline, delivery, cleanup, restoration, and sampled-ledger invariants.

DAP remains stable when explicit action-effect estimates are perturbed. Across 96,000 chronological development decisions, $\pm 5\%$ changes at most 6.91% of actions and 0.56 completion points; $\pm 10\%$ changes 10.55% and 0.62 points. The mask records zero trace-token overspend. This sensitivity characterizes the action-effect calibration used to construct the branches.

6 Discussion and Applicability

The results support a software design principle: preserve explicit action semantics where resource decisions are compared, delivered, and charged. Matched-transition controls connect this structure to service productivity; the PDS adapter demonstrates reusable verification across value representations. An independently witnessed command can reveal a selection discrepancy even when both

targets are locally valid. The separate ledger similarly connects admission to measured consumption. These relations help locate selection, delivery, and accounting discrepancies at their software boundaries.

Applying the contract requires an action-effect model, a resource ledger, and an observable actuator. Trace scheduling uses exact token costs; Kubernetes uses calibrated capacity, recovery timing, and sampled Ready accounting. A new deployment calibrates these quantities while preserving common-context, feasible-selection, and receipt checks. Under the conditions in Section 4, the reserve connects each admitted command to its return to base. Thus the interface transfers with workload- and deployment-specific models and resource semantics.

Each evaluation supplies a distinct form of evidence. Paired trace seeds measure service productivity; exact DP isolates action ordering; and paired HTTP plans measure live service and Ready cost on the tested cluster. Trace SLO measures describe the queue model, while live outcomes use request receipts. Pairing and disjoint selection inputs make mechanism contrasts interpretable within those conditions. Together, these designs separate the quality of a resource decision from its preservation through execution.

7 Related Work

Autonomic control and cloud resource management. Autonomic software frames adaptation as a monitored feedback loop [27]. Cloud controllers combine queue models, forecasting, and autoscaling: AutoControl, AutoScale, Autopilot, Sinan, and Autothrottle cover virtualized capacity and microservice SLOs [18, 34, 36, 50, 53]. Recent FSE systems add knowledge loops, predictive PID, content-aware signals, and LLM-assisted Kubernetes operations [11, 17, 49, 51]. They expose state and commands, whereas DAP records candidate consequences, budget commitment, comparable scores, delivered command, and actuator/ledger evidence. Kubernetes separates desired state, readiness, and reconciliation [9, 44, 45]; DAP makes the resulting boundaries auditable through forecast equality, feasible selection, target mapping, delivered identity, and ledger closure.

Cluster scheduling and learned policies. Cluster and serving systems optimize placement, packing, interference, and tail behavior [3, 4, 13–16, 19–22, 24, 26, 33, 38, 39, 47, 48]. DeepRM and Decima learn policies from interaction, while constrained methods add costs, trust regions, Lyapunov bounds, or penalty updates [1, 5, 12, 29, 30, 40, 43, 52]. Budgeted RL and LCPO address available budgets and changing contexts [10, 23]. DAP complements these policy and value representations with explicit branch provenance, hard affordability, and planner-to-actuator identity.

Planning, post-decision value, and action ordering. Dyna, MPC, and ADP combine learning, constrained search, and value approximation [31, 35, 41]. Post-decision-state methods separate the action-controlled transition from subsequent exogenous uncertainty and can impose structural value properties [8, 25]. DAP carries this decomposition into deployment: every affordable candidate receives one causal realization, an explicit post-cost state and score, and a preserved identity through argmax, actuation, and ledger closure. PDS/ADP and MPC isolate value and search components, while exact DP measures ordering directly.

8 Conclusion

Direct Action Planning makes affordable action comparison an explicit interface between learning and execution. Structured branches combine known resource effects with post-cost continuation, while decision records connect selection to actuation and accounting. Matched controls demonstrate resource productivity and efficient planning; DAP/PDS checks establish reusable verification; and Kubernetes runs demonstrate faithful deployment. Preserving action semantics across these boundaries makes finite-budget scheduling both effective and independently inspectable.

9 Data Availability

The anonymous replication package is available at <https://anonymous.4open.science/r/direct-action-planning-reproducibility-3642/>. It contains source code, frozen checkpoints, trace preprocessing, configurations, controller and environment definitions, and the manuscript's tables and figures. The evidence/ directory provides per-episode/per-plan outcomes, all 66 matched statistical contrasts, selected continuation weights, executable-fault records, 60 lifecycle measurements, runtime versions, checksums, and a script that recomputes the paired analyses. The main text specifies the methods, metrics, experimental conditions, and principal results; the repository provides the corresponding machine-readable evidence and detailed protocols. Azure acquisition and GenTD26's pinned source, preprocessing recipe, and access instructions are documented under data/; redistribution follows the source datasets' terms. A public release will be maintained upon acceptance.

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